

HORNSBY GIRLS HIGH SCHOOL



Mathematics Extension 1

Year 12 Higher School Certificate

Trial Examination Term 3 2020

STUDENT NUMBER: _____

STUDENT NAME: _____

TEACHER NAME: _____

**General
Instructions:**

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11–14, show relevant mathematical reasoning and/ or calculations

Total Marks:
70

Section I – 10 marks (pages 2–6)

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 7–12)

- Attempt Questions 11–14
- Start each question in a new writing booklet
- Write your student number on every writing booklet
- Allow about 1 hour and 45 minutes for this section

Question	1-10	11	12	13	14	Total
Total	/10	/15	/15	/15	/15	/70

This assessment task constitutes 30% of the Higher School Certificate Course School Assessment

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Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1. Let $P(x) = x^2 + bx + c$ where b and c are constants. The zeros of $P(x)$ are α and $\alpha + 1$.

What are the correct expressions for b and c in terms of α ?

- (A) $b = -(2\alpha + 1)$ and $c = \alpha^2 + \alpha$
(B) $b = 2\alpha + 1$ and $c = \alpha^2 + \alpha$
(C) $b = \alpha^2 + \alpha$ and $c = -(2\alpha + 1)$
(D) $b = \alpha^2 + \alpha$ and $c = 2\alpha + 1$

2. What is the derivative of $\tan^{-1}(2x - 1)$?

- (A) $\frac{1}{4x^2 - 4x + 2}$
(B) $\frac{2x - 1}{2x^2 - 2x + 1}$
(C) $\frac{2}{2x^2 - 2x + 1}$
(D) $\frac{1}{2x^2 - 2x + 1}$

3. 80 students are allocated at random into different study groups. Determine the number of groups required so that **at least one group** has **9 or more** students.

- (A) 11
(B) 10
(C) 9
(D) 8

4. Which of the following expressions is equal to $\cos x + \sin x$?

(A) $\sqrt{2} \sin\left(x + \frac{\pi}{4}\right)$

(B) $2 \sin\left(x + \frac{\pi}{4}\right)$

(C) $\sqrt{2} \sin\left(x - \frac{\pi}{4}\right)$

(D) $2 \sin\left(x - \frac{\pi}{4}\right)$

5. Four males and four females are to sit around a table. In how many ways can this be done if the males and females alternate?

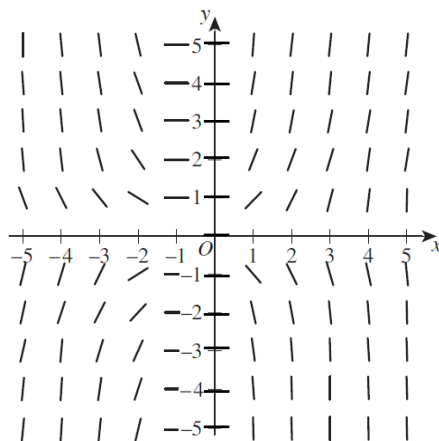
(A) 144

(B) 2880

(C) 5040

(D) 40 320

6. The direction (slope) field for a first order differential equation is shown.



Which of the following could be the differential equation represented?

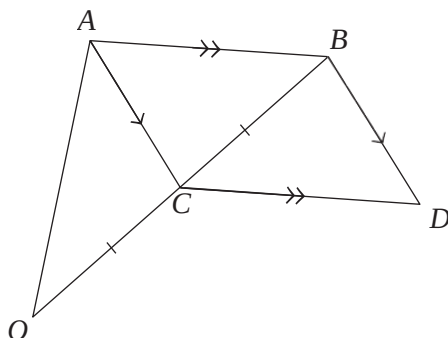
(A) $\frac{dy}{dx} = (x+1)^3$

(B) $\frac{dy}{dx} = x(y+1)$

(C) $\frac{dy}{dx} = (x+1)y$

(D) $\frac{dy}{dx} = (x-1)y$

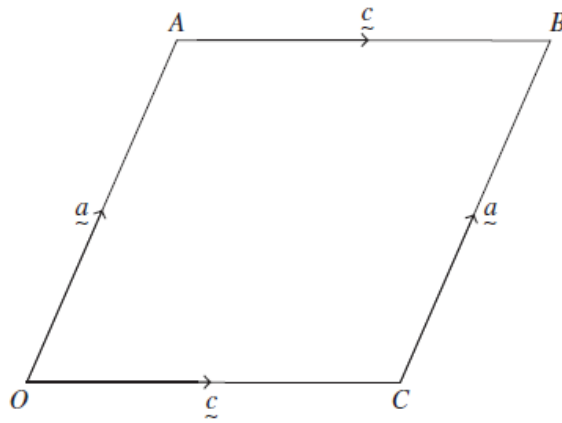
7. The position vectors of points A and B are $\underline{a}$ and $\underline{b}$ respectively. Point C is the midpoint of OB and point D is such that $ABDC$ is a parallelogram.



Which of the following is the position vector of D ?

- (A) $\frac{3}{2}\underline{b} + \underline{a}$
- (B) $\frac{3}{2}\underline{b} - \underline{a}$
- (C) $\frac{1}{2}\underline{b} - \frac{1}{2}\underline{a}$
- (D) $\frac{1}{2}\underline{b} - \underline{a}$
8. What is the value of k such that the function $f(x) = \frac{k}{1+x^2}$, $-1 \leq x \leq 1$ is a probability density function?
- (A) $k = \frac{\pi}{4}$
- (B) $k = \frac{\pi}{2}$
- (C) $k = \frac{2}{\pi}$
- (D) $k = \frac{4}{\pi}$
9. A curve C has parametric equations $x = \cos^2 t$ and $y = 4\sin^2 t$ for $t \in \mathbb{R}$. What is the Cartesian equation of C ?
- (A) $y = 1 - x$ for $0 \leq x \leq 1$
- (B) $y = 4 - 4x$ for $t \in \mathbb{R}$
- (C) $y = 4 - 4x$ for $0 \leq x \leq 1$
- (D) $y = 1 - x$ for $t \in \mathbb{R}$

10. The diagram shows $OABC$, a rhombus in which $\overrightarrow{OA} = \overrightarrow{CB} = \underline{a}$ and $\overrightarrow{OC} = \overrightarrow{AB} = \underline{c}$.



To prove that the diagonals of OABC are perpendicular, it is required to show that

- (A) $(\underline{a} + \underline{c}) \cdot (\underline{a} + \underline{c}) = 0$
- (B) $(\underline{a} - \underline{c}) \cdot (\underline{a} - \underline{c}) = 0$
- (C) $(\underline{a} - \underline{c}) \cdot (\underline{a} + \underline{c}) = 0$
- (D) $\underline{a} \cdot \underline{c} = 0$

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a new writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a new writing booklet

- (a) Consider the function $f(x) = x^2 - 4x + 6$.
- (i) Explain why the domain of $f(x)$ must be restricted if $f(x)$ is to have an inverse function. 1
- (ii) Given that the domain of $f(x)$ is restricted to $x \leq 2$, find an expression for $f^{-1}(x)$. 2
- (iii) Given the restriction in part (a) (ii), state the domain and range of $f^{-1}(x)$. 2
- (iv) The curve $y = f(x)$ ($x \leq 2$) and the curve $y = f^{-1}(x)$ intersect at the point P . 2
- Find the coordinates of P .
- (b) Use the substitution $u = 1 + 2 \tan x$ to evaluate $\int_0^{\frac{\pi}{4}} \frac{1}{(1 + 2 \tan x)^2 \cos^2 x} dx$. 2
- (c) Use t -formulae to solve the equation $\cos x - \sin x = -1$, where $0 \leq x \leq 2\pi$. 3
- (d) Use the substitution $u = 1 - x$ to evaluate $\int_{-3}^0 \frac{x}{\sqrt{1-x}} dx$ 3

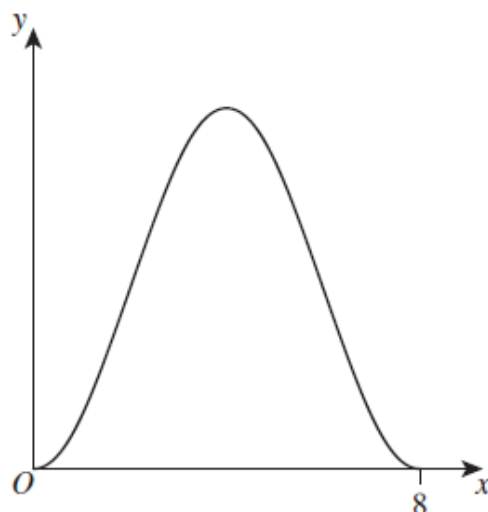
End of Question 11

Question 12 (15 marks) Start a new writing booklet

(a) Evaluate $\int_0^{\frac{2}{3}} \frac{1}{\sqrt{4-9x^2}} dx$.

3

- (b) A proposed plan for a garden is shown in the diagram. The curved boundary of the garden is modelled by the function $f(x) = 6\sin^2\left(\frac{\pi x}{8}\right)$, $0 \leq x \leq 8$.



- (i) Use the identity $\sin A \sin B = \frac{1}{2}[\cos(A-B) - \cos(A+B)]$ to show that

2

$$\sin^2\left(\frac{\pi x}{8}\right) = \frac{1}{2}\left(1 - \cos\frac{\pi x}{4}\right) .$$

- (ii) Use the result from part (a) (i) to find the area A of the garden.

2

Question 12 continues on page 9

Question 12

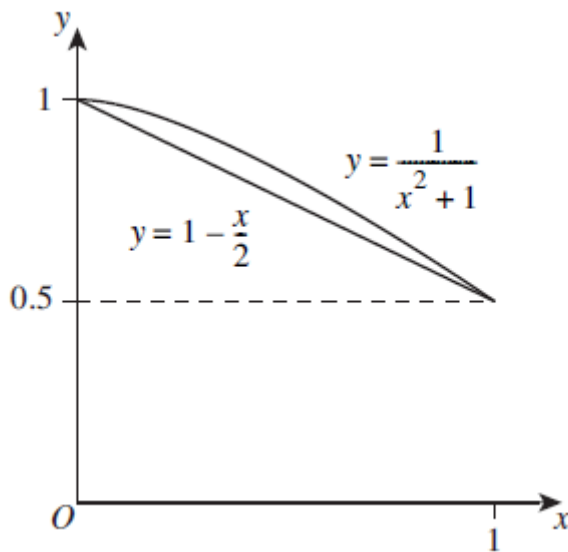
- (c) The work done W , by a constant force $\vec{F}$, in moving a particle through a displacement $\vec{s}$, is defined by the formula $W = \vec{F} \cdot \vec{s}$. A force described by the vector $\vec{F} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$ moves a particle along the line l from $P(-1, 2)$ to $Q(2, -2)$ so that $\vec{s} = \overrightarrow{PQ}$.

(i) Find $\vec{s}$ and hence find the value of W . 1

(ii) Hence, verify that W is also given by $W = (\vec{F} \cdot \hat{s}) |\vec{s}|$. 1

(iii) Find the vector projection of $\vec{F}$ onto $\vec{s}$. 1

- (d) The diagram shows the graph of $y = \frac{1}{x^2 + 1}$ and the graph of $y = 1 - \frac{x}{2}$ for $0 \leq x \leq 1$.



(i) Find the exact volume of the solid of revolution formed when the region bounded by the graph of $y = \frac{1}{x^2 + 1}$, the y -axis and the line $y = \frac{1}{2}$ is rotated 360° about the y -axis. 2

(ii) Find the exact volume of the solid of revolution formed when the region bounded by the graph of $y = 1 - \frac{x}{2}$, the y -axis and the line $y = \frac{1}{2}$ is rotated 360° about the y -axis. 2

(iii) Use the results from parts (c) (i) and (ii) to show that $\ln 2 > \frac{2}{3}$. 1

End of Question 12

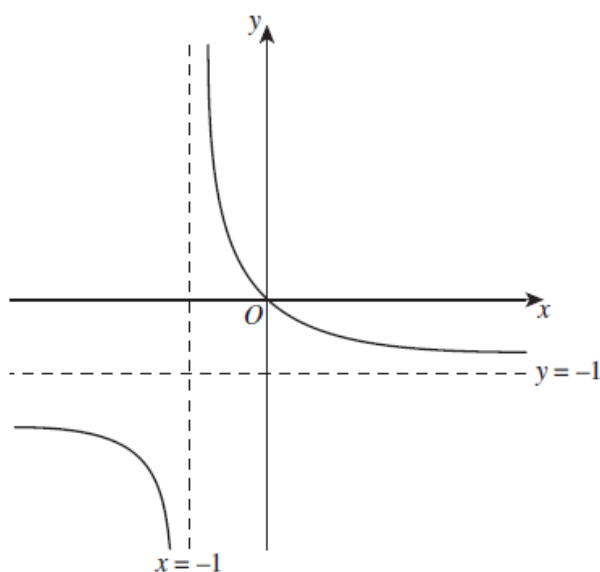
Question 13 (15 marks) Start a new writing booklet

- (a) Prove by mathematical induction that, for all integers $n \geq 1$,

3

$$\frac{2}{1 \times 3} + \frac{2}{2 \times 4} + \frac{2}{3 \times 5} + \dots + \frac{2}{n(n+2)} = \frac{3}{2} - \frac{2n+3}{(n+1)(n+2)} .$$

- (b) The diagram below is a sketch of the graph of the function $f(x) = \frac{-x}{x+1}$.



- (i) Sketch the graph of $y = -f(x)$, showing all asymptotes and intercepts.

2

- (ii) Sketch the graph of $y = (f(x))^2$, showing all asymptotes and intercepts.

2

- (iii) Solve the equation $(f(x))^2 = f(x)$ for x .

1

Question 13 continues on page 11

Question 13

- (c) The area $A \text{ cm}^2$ is occupied by a bacterial colony. The colony has its growth modelled by

The logistic equation $\frac{dA}{dt} = \frac{1}{25} A(50 - A)$ where $t \geq 0$ and t is measured in days.

At time $t = 0$, the area occupied by the bacteria colony is $\frac{1}{2} \text{ cm}^2$.

- (i) Show that $\frac{1}{A(50 - A)} = \frac{1}{50} \left(\frac{1}{A} + \frac{1}{50 - A} \right)$. 1

- (ii) Using the result from part (c) (i), solve the logistic equation and 3

hence show that $A = \frac{50}{1 + 99e^{-2t}}$.

- (iii) According to this model, what is the limiting area of the bacteria colony? 1

- (iv) Find the exact time when the rate of change in the area occupied by the bacterial colony is 2
at its maximum.

End of Question 13

Question 14 (15 marks) Start a new writing booklet

- (a) At time t seconds the length of the side of a cube is x cm, the surface area of the cube is $S \text{ cm}^2$, and the volume of the cube is $V \text{ cm}^3$. The surface area of the cube is increasing at a constant rate of $8 \text{ cm}^2 / \text{s}$.

(i) Show that $\frac{dx}{dt} = \frac{k}{x}$ where k is a constant. 2

(ii) Show that $\frac{dV}{dt} = 2V^{\frac{1}{3}}$. 2

(iii) Given that $V = 8$ when $t = 0$, find the value of t when $V = 16\sqrt{2}$. 2

(b)

(i) Prove the trigonometric identity $\tan 3\theta = \frac{3\tan \theta - \tan^3 \theta}{1 - 3\tan^2 \theta}$. 3

(ii) Use the identity from part (b) (i) to show that the roots of the cubic equation $x^3 - 3x^2 - 3x + 1 = 0$ are $\tan \frac{\pi}{12}$, $\tan \frac{5\pi}{12}$ and -1 . 2

(iii) Hence, show that $\tan \frac{\pi}{12} = 2 - \sqrt{3}$. 2

- (c) The table shows selected values of a one-to-one differentiable function $g(x)$ and its derivative $g'(x)$. 2

x	-1	0
$g(x)$	-5	-1
$g'(x)$	3	$\frac{1}{2}$

Let $f(x)$ be a function such that $f(x) = g^{-1}(x)$. Find the value of $f'(-1)$.

End of paper

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Hornsby Girls High School
Year 12 Mathematics Extension 1 HSC Trial 2020
Solutions

Solutions	Marker's Comments
Question 1 A If α and $\alpha + 1$ are zeros of $P(x)$, then $P(x) = x^2 - (2\alpha + 1)x + (\alpha^2 + \alpha).$ Equating coefficients gives $b = -(2\alpha + 1)$ and $c = \alpha^2 + \alpha$.	$\frac{52}{69}$ were correct.
Question 2 D $\frac{d}{dx}(\tan^{-1} f(x)) = \frac{f'(x)}{1 + (f(x))^2}$ $f(x) = 2x - 1$ and $f'(x) = 2$ $\begin{aligned} \frac{d}{dx}(\tan^{-1}(2x - 1)) &= \frac{2}{1 + (2x - 1)^2} \\ &= \frac{2}{4x^2 - 4x + 2} \\ &= \frac{1}{2x^2 - 2x + 1} \end{aligned}$	$\frac{54}{69}$ were correct.
Question 3 C Since there are 80 students, they could be divided into 10 groups each with 8 students. To ensure that at least one group has 9 students, we need 9 groups.	$\frac{46}{69}$ were correct.
Question 4 A $R \sin(x + \alpha) = R \sin x \cos \alpha + R \cos x \sin \alpha$ $\sin x + \cos x = R \sin x \cos \alpha + R \cos x \sin \alpha$ Equating coefficients of $\sin x$ gives $R \cos \alpha = 1$. (1) Equating coefficients of $\cos x$ gives $R \sin \alpha = 1$. (2) Squaring both (1) and (2) and adding gives $R^2 = 2 \Rightarrow R = \sqrt{2} (> 0)$. Substituting into (1) and (2) gives $\cos \alpha = \frac{1}{\sqrt{2}}$ and $\sin \alpha = \frac{1}{\sqrt{2}}$. So $\alpha = \frac{\pi}{4}$ and hence $\sin x + \cos x = \sqrt{2} \sin\left(x + \frac{\pi}{4}\right)$.	$\frac{56}{69}$ were correct.

Solutions	Marker's Comments																
Question 5 A The table outlines the possible seating arrangements. <table><tr><td>M1</td><td>M2</td><td>M3</td><td>M4</td><td>F1</td><td>F2</td><td>F3</td><td>F4</td></tr><tr><td>1</td><td>3</td><td>2</td><td>1</td><td>4</td><td>3</td><td>2</td><td>1</td></tr></table> Therefore the number of possible seating arrangements is $3! \times 4! = 144$.	M1	M2	M3	M4	F1	F2	F3	F4	1	3	2	1	4	3	2	1	$\frac{55}{69}$ were correct.
M1	M2	M3	M4	F1	F2	F3	F4										
1	3	2	1	4	3	2	1										
Question 6 C At $(0, 0)$, $\frac{dy}{dx} = 0$ and so A is incorrect. At $(-1, 1)$, $\frac{dy}{dx} = 0$ and so B and D are incorrect.	$\frac{42}{69}$ were correct. Many chose B wrongly.																
Question 7 B $\begin{aligned}\overrightarrow{OD} &= \overrightarrow{OC} + \overrightarrow{CD} \\ &= \frac{1}{2}\overrightarrow{OB} + \overrightarrow{AB} \\ &= \frac{1}{2}\overrightarrow{OB} + \overrightarrow{AO} + \overrightarrow{OB} \\ &= \frac{1}{2}\underline{b} - \underline{a} + \underline{b} \\ &= \frac{3}{2}\underline{b} - \underline{a}\end{aligned}$	$\frac{57}{69}$ were correct.																
Question 8 C $\begin{aligned}\int_{-1}^1 \frac{k}{1+x^2} \, dx &= 1 \\ k(\tan^{-1} 1 - \tan^{-1}(-1)) &= 1 \\ k\left(\frac{\pi}{4} + \frac{\pi}{4}\right) &= 1 \\ k &= \frac{2}{\pi}\end{aligned}$	$\frac{46}{69}$ were correct.																

Solutions	Marker's Comments
Question 9 C The parametric equations are: $x = \cos^2 t \quad (1)$ $y = 4 \sin^2 t \quad (2)$ $\frac{(2)}{4} \text{ gives } \frac{y}{4} = \sin^2 t. \quad (3)$ (1) + (3) and using $\cos^2 t + \sin^2 t = 1$ gives $x + \frac{y}{4} = 1 \Rightarrow 4x + y = 4$. $0 \leq \cos^2 t \leq 1$ and so $0 \leq x \leq 1$. Therefore, $y = 4 - 4x$ for $0 \leq x \leq 1$.	$\frac{41}{69}$ were correct. Many chose B wrongly
Question 10 C The diagonals of $OABC$ are given by $\overrightarrow{OB}$ and $\overrightarrow{CA}$. To prove they are perpendicular, form $\overrightarrow{CA} \cdot \overrightarrow{OB}$ and show that it equals zero. $\overrightarrow{CA} = \underline{a} - \underline{c}$ and $\overrightarrow{OB} = \underline{a} + \underline{c}$. Therefore $\overrightarrow{CA} \cdot \overrightarrow{OB} = 0$ if $(\underline{a} - \underline{c}) \cdot (\underline{a} + \underline{c}) = 0$.	$\frac{58}{69}$ were correct.
Question 11 (a) (i) $f(x) = x^2 - 4x + 6$ is a parabola. Excluding the turning point at (2, 2), for each value of $f(x)$ in the range there are two x-values. Geometrically, this corresponds to a horizontal line intersecting the graph twice. If x and y are swapped, each x-value in the domain will have two y-values. Hence the inverse will not be a function.	1 mark Many students mistakenly thought that $f^{-1}(x)$ meant $(f(x))^{-1} = \frac{1}{f(x)}$ then gave a reason of restricting the domain to not include dividing by zero
(ii) Use the completing the square method to express $f(x)$ in turning point form: $f(x) = x^2 - 4x + 6 \quad (x \leq 2)$ $= (x - 2)^2 + 2$ Swap x and y, then make y the subject. $x = (y - 2)^2 + 2$ $x - 2 = (y - 2)^2$ $y - 2 = -\sqrt{x - 2} \quad (\sqrt{x - 2} \text{ is discarded as } y \leq 2)$ $y = -\sqrt{x - 2} + 2$ $f^{-1}(x) = -\sqrt{x - 2} + 2$	1 for interchanging x and y 1 for correct equation

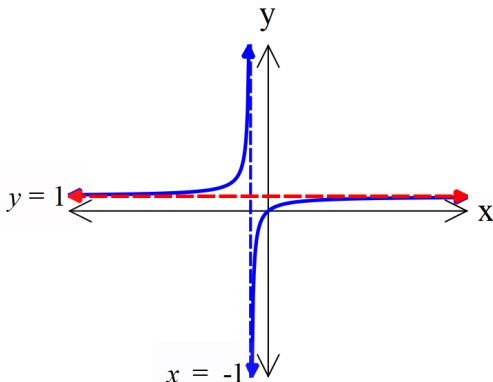
Solutions	Marker's Comments
Question 11 (a) (iii) The domain is $x \geq 2$ as $x - 2 \geq 0$. The range is $y \leq 2$ as $-\sqrt{x-2} \leq 0$.	2 marks
(iv) The curves $y = f(x)$ and $y = f^{-1}(x)$ have a common intersection with the line $y = x$. For example, attempting to solve $f(x) = x$ for x: $x^2 - 4x + 6 = x$ $x^2 - 5x + 6 = 0$ $x = 2, 3$ When $x = 2$, $y = 2$ and so $(2, 2)$ lies on the line $y = x$. When $x = 3$, $y = 1$ and so $(3, 1)$ does not lie on the line $y = x$. Therefore the coordinates of P are $(2, 2)$.	2 marks
(b) Let $u = 1 + 2 \tan x$. $\frac{du}{dx} = 2 \sec^2 x = \frac{2}{\cos^2 x} \Rightarrow dx = \frac{\cos^2 x}{2} du$ When $x = 0$, $u = 1$ and when $x = \frac{\pi}{4}$, $u = 3$. $\int_0^{\frac{\pi}{4}} \frac{1}{(1 + 2 \tan x)^2 \cos^2 x} dx = \int_1^3 \frac{1}{2u^2} du$ $= -\left[\frac{1}{2u}\right]_1^3$ $= -\left(\frac{1}{6} - \frac{1}{2}\right)$ $= \frac{1}{3}$	Generally well done. Some careless mistakes.
(c) substitute $t = \tan \frac{x}{2}$, $\frac{1-t^2}{1+t^2} - \frac{2t}{1+t^2} = -1$ $1-t^2-2t = -1-t^2$ $2t = 2$ $t = 1$ $\tan \frac{x}{2} = 1, \quad 0 \leq x \leq 2\pi$ $\frac{x}{2} = \frac{\pi}{4}, \quad 0 \leq \frac{x}{2} \leq \pi$ $x = \frac{\pi}{2}$ Also when $x = \pi$, $\cos x - \sin x = -1 - 0 = -1$ $\therefore x = \frac{\pi}{2}, \pi$	Most students used t-formula and find t correctly. Some students didn't check the domain and find the correct number of solutions in the domain. Some forgot to check the solution of π.

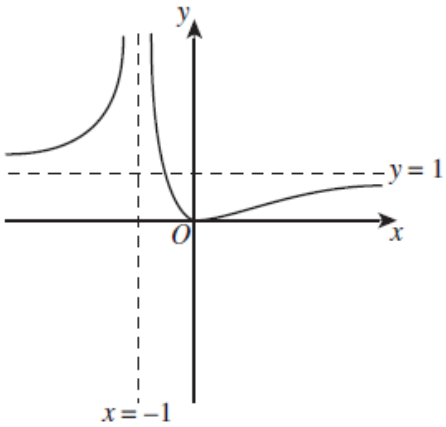
Solutions	Marker's Comments
Question 11 (d) $u = 1 - x$ $\frac{du}{dx} = -1 \therefore dx = -du$ when $x = -3, u = 4$ when $x = 0, u = 1$ $\int_{-3}^0 \frac{x}{\sqrt{1-x}} dx = \int_1^4 \frac{1-u}{\sqrt{u}} (-du)$ $= \int_1^4 u^{-\frac{1}{2}} - u^{\frac{1}{2}} du$ $= \left[2u^{\frac{1}{2}} - \frac{2}{3}u^{\frac{3}{2}} \right]_1^4$ $= \left(4 - \frac{16}{3} \right) - \left(2 - \frac{2}{3} \right)$ $= -\frac{4}{3} - \frac{4}{3}$ $= -\frac{8}{3}$	Generally well done. Some careless mistakes.
Question 12 (a) $F(x) = \int_0^{\frac{2}{3}} \frac{1}{\sqrt{4-9x^2}} dx$ $= \frac{1}{3} \left[\sin^{-1} \left(\frac{3}{2}x \right) \right]_0^{\frac{2}{3}}$ $= \frac{1}{3} [\sin^{-1} 1 - \sin^{-1} 0]$ $= \frac{1}{3} \times \frac{\pi}{2}$ $= \frac{\pi}{6}$ Or	1 for recognising and manipulating to a form which can be integrated 1 for correct integration 1 for substitution and correct evaluation

$ \begin{aligned} F(x) &= \int_0^{\frac{2}{3}} \frac{1}{\sqrt{4-9x^2}} dx \\ &= \int_0^{\frac{2}{3}} \frac{1}{\sqrt{9(\frac{4}{9}-x^2)}} dx \\ &= \int_0^{\frac{2}{3}} \frac{1}{3\sqrt{(\frac{4}{9}-x^2)}} dx \\ &= \frac{1}{3} \left[\sin^{-1} \left(\frac{x}{\frac{2}{3}} \right) \right]_0^{\frac{2}{3}} \\ &= \frac{1}{3} \left[\sin^{-1} \left(\frac{3}{2}x \right) \right]_0^{\frac{2}{3}} \\ &= \frac{1}{3} [\sin^{-1} 1 - \sin^{-1} 0] \\ &= \frac{1}{3} \times \frac{\pi}{2} \\ &= \frac{\pi}{6} \end{aligned} $	
(b) (i) Using $\sin A \sin B = \frac{1}{2} [\cos(A-B) - \cos(A+B)]$ with $A = B = \frac{\pi x}{8}$ gives: LHS = $\sin \frac{\pi x}{8} \sin \frac{\pi x}{8}$ $= \sin^2 \frac{\pi x}{8}$ RHS = $\frac{1}{2} \left[\cos \left(\frac{\pi x}{8} - \frac{\pi x}{8} \right) - \cos \left(\frac{\pi x}{8} + \frac{\pi x}{8} \right) \right]$ $= \frac{1}{2} \left(\cos 0 - \cos \frac{\pi x}{4} \right)$ $= \frac{1}{2} \left(1 - \cos \frac{\pi x}{4} \right)$ So $\sin^2 \left(\frac{\pi x}{8} \right) = \frac{1}{2} \left(1 - \cos \frac{\pi x}{4} \right)$.	1 for correct rearrangement 1 for evaluation and simplifying Students need to show all working in show that questions eg $\cos 0$ should be included in the working here
Solutions	Marker's Comments
Question 12 (b) (ii) $A = 6 \int_0^8 \sin^2 \left(\frac{\pi x}{8} \right) dx$ $= 3 \int_0^8 1 - \cos \frac{\pi x}{4} dx$ $= 3 \left[x - \frac{4}{\pi} \sin \frac{\pi x}{4} \right]_0^8$ $= 3 \left(8 - \frac{4}{\pi} \sin 2\pi - (0 - \sin 0) \right)$ $= 3(8 - 0)$ $= 24 \text{ unit}^2$	Many students missed the 6 1 for rearrangement and integration 1 for substitution and correct evaluation

(c) (i) Substituting $\underline{F} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$ and $\underline{s} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ into $W = \underline{F} \cdot \underline{s} \text{ gives:}$ $W = \underline{F} \cdot \underline{s}$ $= \begin{pmatrix} 4 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ $= 20$	Part c was done well by most students 1
(ii) A unit vector in the direction of $\overrightarrow{PQ}$ is $\hat{\underline{s}} = \frac{1}{5} \begin{pmatrix} 3 \\ -4 \end{pmatrix}$. Substituting $\underline{F} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$, $\hat{\underline{s}} = \frac{1}{5} \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ and $\underline{s} = 5$ into $W = (\underline{F} \cdot \hat{\underline{s}}) \underline{s} \text{ gives:}$ $W = \left(\begin{pmatrix} 4 \\ -2 \end{pmatrix} \cdot \frac{1}{5} \begin{pmatrix} 3 \\ -4 \end{pmatrix} \right) 5$ $= 20$	1
(iii) The component of $\underline{F}$ in the direction of l is given by $\left(\frac{\underline{F} \cdot \underline{s}}{\underline{s} \cdot \underline{s}} \right) \underline{s}.$ Substituting $\underline{F} \cdot \underline{s} = 20$, $\underline{s} \cdot \underline{s} = 25$ and $\underline{s} = \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ into $\left(\frac{\underline{F} \cdot \underline{s}}{\underline{s} \cdot \underline{s}} \right) \underline{s} \text{ gives:}$ $\left(\frac{\underline{F} \cdot \underline{s}}{\underline{s} \cdot \underline{s}} \right) \underline{s} = \frac{20}{25} \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ $= \frac{4}{5} \begin{pmatrix} 3 \\ -4 \end{pmatrix}$ $= \begin{pmatrix} 2.4 \\ -3.2 \end{pmatrix}$ Alternatively, the component of $\underline{F}$ in the direction of l is $(\underline{F} \cdot \hat{\underline{s}})\hat{\underline{s}}$.	1 Some students had a scalar answer instead of a vector
Solutions	Marker's Comments
Question 12 (d) (i) Rearranging $y = \frac{1}{x^2 + 1}$ to express x^2 in terms of y gives $x^2 = \frac{1}{y} - 1$. $V = \pi \int_{\frac{1}{2}}^1 \left(\frac{1}{y} - 1 \right) dy$ $= \pi \left[\ln y - y \right]_{\frac{1}{2}}^1$ $= \pi \left(\ln 1 - 1 - \left(\ln \frac{1}{2} - \frac{1}{2} \right) \right)$ $= \pi \left(\ln 2 - \frac{1}{2} \right)$	When finding a volume about the y axis we need to use the form $V = \pi \int x^2 dy$ not $V = \pi \int y^2 dx$ Some students had trouble finding the correct expression for x^2 1 for correct integral statement 1 for correct integration and evaluation

(ii) Rearranging $y = 1 - \frac{x}{2}$ to express x in terms of y gives $x = 2(1 - y).$ $V = \pi \int_{\frac{1}{2}}^1 (4(1 - y)^2) dy$ $= -\frac{4\pi}{3} \left[(1 - y)^3 \right]_{\frac{1}{2}}^1$ $= -\frac{4\pi}{3} \left(0 - \frac{1}{8} \right)$ $= \frac{\pi}{6}$ Alternatively: The solid formed is a cone of radius 1 and height $\frac{1}{2}$. Substituting these values into $V = \frac{1}{3} \pi r^2 h$ gives: $V = \frac{1}{3} \times \pi \times 1^2 \times \frac{1}{2}$ $= \frac{\pi}{6}$	Similar problems as part i 1 for correct integral statement 1 for correct integration and evaluation No-one used this approach
The volume of the solid under the curve $y = \frac{1}{x^2 + 1}$ is greater than the volume of the solid under the curve $y = 1 - \frac{x}{2}$, $\therefore V(i) > V(ii)$ (iii) From the diagram, it can be reasoned that $\pi \left(\ln 2 - \frac{1}{2} \right) > \frac{\pi}{6}.$ So $\ln 2 - \frac{1}{2} > \frac{1}{6} \Rightarrow \ln 2 > \frac{2}{3}.$	1

Solutions	Marker's Comments
Question 13 (a) Consider $n = 1$. $\text{LHS} = \frac{2}{1 \times 3} = \frac{2}{3} \text{ and}$ $\text{RHS} = \frac{3}{2} - \frac{2(1)+3}{(1+1)(1+2)} = \frac{4}{6} = \frac{2}{3} = \text{LHS.}$ The statement is true when $n = 1$. Suppose true for $n = k$. $\text{So } \frac{2}{1 \times 3} + \frac{2}{2 \times 4} + \frac{2}{3 \times 5} + \dots + \frac{2}{k(k+2)} = \frac{3}{2} - \frac{2k+3}{(k+1)(k+2)}.$ Show it is true for $n = k + 1$; that is, $\frac{2}{1 \times 3} + \frac{2}{2 \times 4} + \frac{2}{3 \times 5} + \dots + \frac{2}{k(k+2)} + \frac{2}{(k+1)(k+3)} =$ $\frac{3}{2} - \frac{2(k+1)+3}{((k+1)+1)((k+1)+2)}$ $\text{LHS} = \frac{2}{1 \times 3} + \frac{2}{2 \times 4} + \frac{2}{3 \times 5} + \dots + \frac{2}{k(k+2)} + \frac{2}{(k+1)(k+3)}$ $= \frac{3}{2} - \frac{2k+3}{(k+1)(k+2)} + \frac{2}{(k+1)(k+3)}$ $= \frac{3}{2} - \frac{(2k+3)(k+3) - 2(k+2)}{(k+1)(k+2)(k+3)}$ $= \frac{3}{2} - \frac{2k^2 + 7k + 5}{(k+1)(k+2)(k+3)}$ $= \frac{3}{2} - \frac{(2k+5)(k+1)}{(k+1)(k+2)(k+3)}$ $= \frac{3}{2} - \frac{2k+5}{(k+2)(k+3)}$ $= \frac{3}{2} - \frac{2(k+1)+3}{((k+1)+1)((k+1)+2)}$ $= \text{RHS}$ If true for $n = k$, then true for $n = k + 1$. Hence, by mathematical induction, true for $n \geq 1$.	Most students followed the proving steps well and wrote the statement correctly. Some made mistake in the algebra and end up “fudging” the result.
(b) (i) 	Most students were able to draw the correct asymptotes and shape of the curve.

Solutions	Marker's Comments
Question 13 (b) (ii) 	Some students didn't recognise that this function is positive for all x. Some didn't find the correct asymptotes.
(iii) $(f(x))^2 = f(x) \Rightarrow f(x)(f(x) - 1) = 0$ So $f(x) = 1$ or $f(x) = 0$. $-\frac{x}{x+1} = 1 \Rightarrow x = -\frac{1}{2}$ Hence $x = -\frac{1}{2}$ or $x = 0$. OR The graphs of $y = f(x)$ and $y = (f(x))^2$ intersect at O, where $x = 0$. The graphs of $y = f(x)$ and $y = (f(x))^2$ intersect on the line $y = 1$, where $x = -\frac{1}{2}$.	Some students missed the solution $x = 0$. Some students include the solution $x = -1$ which is not in the domain.
(c) (i) Start with the RHS and show that it equals the LHS. $\begin{aligned} \text{RHS} &= \frac{1}{50} \left(\frac{(50-A)+A}{A(50-A)} \right) \\ &= \frac{1}{A(50-A)} \\ &= \text{LHS} \end{aligned}$	Mostly well done.

Solutions	Marker's Comments
Question 13 (c) (ii) This is a differential equation of the form $\frac{dA}{dt} = g(A)$. Attempt to separate variables and integrate both sides. $\int 1 dt = \int \frac{25}{A(50 - A)} dA$ $t = \frac{1}{2} \int \left(\frac{1}{A} + \frac{1}{50 - A} \right) dA \text{ (using the part (i) result)}$ $= \frac{1}{2} (\ln A - \ln 50 - A) + c$ $= \frac{1}{2} \ln \left \frac{A}{50 - A} \right + c$ Rearranging gives $A_0 e^{2t} = \frac{A}{50 - A}$ where $A_0 = e^{-2c}$ and hence $A_0 > 0$. When $t = 0$, $A = \frac{1}{2}$ and so $A_0 = \frac{1}{99}$. <i>Note: There are various possible ways to find the value of the constant.</i> $e^{2t} = \frac{99A}{50 - A}$ $99Ae^{-2t} = 50 - A$ $A(1 + 99e^{-2t}) = 50$ So $A = \frac{50}{1 + 99e^{-2t}}$.	Generally well done. Some mistakes solving the DE and finding the constant.
(iii) As $t \rightarrow \infty$, $1 + 99e^{-2t} \rightarrow 1$ and so $A \rightarrow \frac{50}{1} = 50$. The limiting area of the bacteria colony is 50 cm^2.	Generally well done.

Solutions	Marker's Comments
(c) (iv) The graph of $\frac{dA}{dt}$ versus A (inverted parabola) has a maximum at $A = 25$. It requires us to find the value of t such that $25 = \frac{50}{1 + 99e^{-2t}}.$ $25(1 + 99e^{-2t}) = 50$ $1 + 99e^{-2t} = 2$ $e^{-2t} = \frac{1}{99}$ $e^{2t} = 99$ $t = \frac{1}{2} \ln 99 \text{ (days)}$ The rate of change of the area is at its maximum at $t = \frac{1}{2} \ln 99$ (days). <i>Note: There are other valid but more time-consuming methods of determining this solution.</i> <i>Method 1:</i> Finding $\frac{d^2A}{dt^2} = \frac{1}{25^2} A(50 - A)(50 - 2A)$, determining that $\frac{dA}{dt}$ is a maximum when $A = 25$ and then solving for t as above. <i>Method 2:</i> Determining the value of t when the (non-stationary) point of inflection occurs by finding $\frac{d^2A}{dt^2}$ in terms of t and then finding the value of t such that $\frac{d^2A}{dt^2} = 0$.	Generally well done. Mark is not granted for carry-forward error trying to solve $e^{2t} = 0$, $\therefore t = 0$

Solutions	Marker's Comments
Question 14 (a) (i) Surface area of a cube with a side length of x is $6x^2$ $S = 6x^2$ $\frac{dS}{dx} = 12x$ Given $\frac{dS}{dt} = 8$ $\frac{dx}{dt} = \frac{dS}{dt} \div \frac{dS}{dx} = \frac{8}{12x}$ $= \frac{2}{3x} \text{ with } k = \frac{2}{3}$	Some careless mistake of $S = x^2$. “Show” that $\frac{dx}{dt} = \frac{k}{x}$: should express in that format.
(ii) $V = x^3$ (volume of a cube with side length x) $\frac{dV}{dx} = 3x^2$ $\frac{dV}{dt} = \frac{dV}{dx} \times \frac{dx}{dt}$ $= 3x^2 \times \frac{2}{3x} = 2x$ $x = \frac{1}{2} \frac{dV}{dt}$ $x = V^{\frac{1}{3}}$ $\frac{1}{2} \frac{dV}{dt} = V^{\frac{1}{3}}$ $\frac{dV}{dt} = 2V^{\frac{1}{3}}$	Many had problem in obtaining the required expression.
(iii) $\frac{dV}{dt} = 2V^{\frac{1}{3}}$ $V^{-\frac{1}{3}} dV = 2 dt$ $\int V^{-\frac{1}{3}} dV = \int 2 dt$ $\frac{3}{2} V^{\frac{2}{3}} = 2t + C$ Given that $V = 8$ when $t = 0$ $\frac{3}{2} 8^{\frac{2}{3}} = 2 \times 0 + C \text{ or } C = 6$ $\frac{3}{2} V^{\frac{2}{3}} = 2t + 6$ Find t when $V = 16\sqrt{2}$ $\frac{3}{2} (16\sqrt{2})^{\frac{2}{3}} = 2t + 6$ $12 = 2t + 6$ $t = 3$	Many had problem with evaluating $(16\sqrt{2})^{\frac{2}{3}}$.

Solutions	Marker's Comments
Question 14 (b) (i) Use of $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$ with $A = B = \theta$. $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$ Use of $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$ with $A = 2\theta$ and $B = \theta$. $\begin{aligned} \tan 3\theta &= \frac{\tan 2\theta + \tan \theta}{1 - \tan 2\theta \tan \theta} \\ &= \frac{\frac{2 \tan \theta}{1 - \tan^2 \theta} + \tan \theta}{1 - \frac{2 \tan \theta}{1 - \tan^2 \theta} \tan \theta} \\ &= \frac{2 \tan \theta + \tan \theta (1 - \tan^2 \theta)}{1 - \tan^2 \theta} \\ &= \frac{(1 - \tan^2 \theta) - 2 \tan^2 \theta}{1 - \tan^2 \theta} \\ &= \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \end{aligned}$	Generally done well. Students need to know the formula for $\tan 2\theta$. A few did not complete their working to the required form.
(ii) Consider $x^3 - 3x^2 - 3x + 1 = 0$ with $x = \tan \theta$. $\tan^3 \theta - 3 \tan^2 \theta - 3 \tan \theta + 1 = 0$ $\frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} = 1$ So $\tan 3\theta = 1$ and finding the roots of $\tan 3\theta = 1$ corresponds to finding the roots of the cubic equation where $x = \tan \theta$. $3\theta = \tan^{-1} 1 + k\pi \text{ where } k \text{ is an integer}$ $\begin{aligned} \theta &= \frac{\pi}{12} + \frac{k\pi}{3} \\ &= \frac{\pi}{12}, \frac{5\pi}{12}, \frac{9\pi}{12} \end{aligned}$	Many did not relate the similarity in structure of $x^3 - 3x^2 - 3x + 1 = 0$ to $\frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$. Many did not connect that $x^3 - 3x^2 - 3x + 1 = 0$ has three roots, neither did they realise that $\tan 3\theta$ indicates there are 3 roots to the solution.

Solutions	Marker's Comments
Question 14 (b) (iii) $\tan \frac{3\pi}{4} = -1$ and so one factor of the cube is $x + 1$. So $x^3 - 3x^2 - 3x + 1 = (x + 1)(x^2 - 4x + 1)$. So $\tan \frac{\pi}{12}$ and $\tan \frac{5\pi}{12}$ are the roots of $x^2 - 4x + 1 = 0$. Solving the quadratic equation $x^2 - 4x + 1 = 0$ for x gives $x = 2 \pm \sqrt{3}$. Since $\tan \frac{\pi}{12} < \tan \frac{5\pi}{12}$, $\tan \frac{\pi}{12}$ is the smaller root and $x = 2 - \sqrt{3}$.	Students have not stated succinctly for the rejection of $2 + \sqrt{3}$ as a solution for $\tan \frac{\pi}{12}$.
(c) From the table, $f(x) = g^{-1}(x)$ and so $f(-1) = g^{-1}(-1) = 0$. $f'(-1) = \frac{1}{g'(f(-1))}$ $= \frac{1}{g'(0)}$ $= \frac{1}{\frac{1}{2}}$ $= 2$	This question is very poorly executed. It is a pity the lack of using graph as a tool to solve this.

End of Paper